SPEAKER 01 / 03 / THE ORIGINAL PAPER AND OUR RECREATION

Ali Omar

Spoken section - approximately 2 minutes

SHOWCASE CUE
Showcase sections: The paper that motivates this project; Same idea, different system; The initial cascade had little time to save.

Good morning. Our project begins with the paper Blended RAG: Improving RAG Accuracy with Semantic Search and Hybrid Query-Based Retrievers, by Sawarkar, Mangal, and Solanki. Their system combines keyword search, dense retrieval, and learned sparse retrieval in Elasticsearch, using ELSER for the sparse branch. The paper studies retrieval and answer quality. Our project asks a narrower question: can we avoid running every retrieval method for every query, while keeping recall close to a blended reference?

We first recreated the broad retrieval recipe on datasets that overlap with the paper. Our Python version uses BM25, MiniLM for dense retrieval, SPLADE for learned sparse retrieval, and reciprocal rank fusion to combine the results. BM25 matches query terms, MiniLM represents semantic similarity, and SPLADE adds learned sparse term weights; they can retrieve different evidence, which motivates combining them. That is a methodological reproduction, not an exact reproduction of the published system. We did not use Elasticsearch or ELSER, and we do not compare our timings directly with the paper's numbers.

The measurements also differ. The original paper reports retrieval metrics and generated-answer scores such as exact match and F1. Our completed study focuses on retrieval latency, Recall@100, and ranking measures. We used the full set of 50 TREC-COVID queries. For the indexed SQuAD follow-up, we used a fixed sample of 256 questions against the full document corpus.

The first cascade taught us why the order of work matters. BM25 took about 1,410 milliseconds out of the 1,493 milliseconds for full fusion on TREC-COVID. A gate cannot save much if the first step already consumes nearly all the time. That became the starting point for our implementation.

275 spoken words

SPEAKER 02 / 03 / OUR CONTRIBUTION AND IMPLEMENTATION

Abdallah Mestrah

Spoken section - approximately 2 minutes

SHOWCASE CUE
Showcase sections: First fix the lexical bottleneck; Indexed BM25; BM25 first, then a learned gate; TREC-COVID results.

Our contribution is an engineering study of two changes: indexed BM25 and a BM25-first decision gate. We are not proposing a new retrieval algorithm. We wanted to measure the indexing gain separately from the extra gain that comes from deciding when to run the neural retrievers.

Every query first runs BM25. The gate then looks at eleven features from the query and its BM25 results. A standardized Ridge regression model predicts how much recall the full fusion system might add. If the predicted gain is small, the system returns BM25. If it is larger, the system runs MiniLM and SPLADE, then fuses those results with the BM25 ranking already in hand. The cascade reuses the BM25 ranking during fallback, so it does not repeat that first search. The model was trained on NFCorpus training queries; relevance labels are not available to the gate at inference time. The gate thresholds came from NFCorpus calibration and stayed fixed, though the indexed TREC primary was selected after quality results had been inspected.

We used two references to keep the comparisons clear. Earlier, non-indexed fusion shows the combined change across separate runs. Equally indexed fusion uses the same BM25 backend as the cascade, so that comparison measures the additional routing gain.

On all 50 TREC-COVID queries, the selected r99 cascade averaged 73.90 milliseconds, compared with 91.28 milliseconds for indexed fusion. That is 1.24 times faster, while retaining 96.68 percent of fusion's mean Recall@100. Compared with our earlier fusion run, it is 20.21 times faster, but that figure includes the indexed BM25 improvement as well as routing. Indexed fusion alone showed a 16.36 times cross-run ratio. These are measured configuration comparisons in our implementation.

279 spoken words

SPEAKER 03 / 03 / RESULTS, LIMITS, AND WHAT COMES NEXT

Abbas Yazbek

Spoken section - approximately 2 minutes

SHOWCASE CUE
Showcase sections: SQuAD; The cascade is not always the best choice; What the evidence supports; Remaining evaluation.

The SQuAD result uses the same full document corpus, but only a declared sample of 256 questions for the indexed follow-up. The fastest tested cascade, r90, averaged 18.59 milliseconds; indexed fusion averaged 29.55 milliseconds. That is 1.59 times faster, with 99.22 percent of fusion's mean Recall@100 on the sample. Against the earlier fusion run on those same questions, the combined ratio is 7.91 times. It includes both indexing and routing.

The cascade is not the best choice in every comparison. On TREC-COVID, standalone SPLADE is faster than indexed fusion and has higher mean recall. On the SQuAD sample, BM25 alone is much faster and retains 95.70 percent of fusion recall, while SPLADE leads the ranking measures. A gain over fusion does not make a cascade the best quality-time option.

Our results differ because our engine, models, corpus, metrics, and timing setup differ; we did not isolate each cause. The TREC primary followed inspection of its quality results; the SQuAD follow-up samples 256 queries. We used one GPU and did not test answer generation or Elasticsearch and ELSER.

Next, we would implement the changes in that original stack, set the evaluation protocol before looking at results, and test answer quality, more datasets, and strong standalone and adaptive baselines. We should also measure index memory, update costs, and repeated timings under different hardware and load.

Our conclusion is modest. Indexed BM25 removed much of the lexical bottleneck in our implementation, and conditional routing added a measurable gain over equally indexed fusion. Whether the same gains transfer to the original Blended RAG system remains an open experiment. Thank you.

266 spoken words